COHEN-MACAULAYNESS IS INHERITED BY A SLICE CUT OUT BY A REGULAR ELEMENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6357-6376 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6378-6434, the complete author proof (55 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author's proof cites EGA, so the card carries a bibliography fragment. That fragment reproduces, field for field and without editing, the @ARTICLE{EGA} entry from the Stacks Project snapshot bibliography shipped with this corpus at focused-seeds/AGP/sources/stacks-snapshot-20261011/my.bib (same upstream Stacks Project snapshot). No entry text was altered and no citation was reconstructed from memory.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms, regular sequences and depth, and nonzerodivisors on fibres; the transfer is retained.

Source reference. Original lemma algebra-lemma-reformulate-CM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma divisors-lemma-composition-regular-immersion: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma divisors-lemma-relative-Cartier: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-slice-once: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6337–6345. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-closed-point-fibre-locally-finite-type: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans;

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-dimension-fibre-at-a-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-locally-quasi-finite-rel-dimension-0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-quasi-finite-points-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma schemes-lemma-fibre-topological: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6434–6434. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, IV Proposition 17.16.1] *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ be a point with image $s \in S$. Assume*

(1) *f is locally of finite presentation,*
(2) *f is Cohen-Macaulay at x, and*
(3) *x is a closed point of X_s.*

Then there exists a regular immersion $Z \to X$ containing x such that

(a) *$Z \to S$ is flat and locally of finite presentation,*
(b) *$Z \to S$ is locally quasi-finite, and*
(c) *$Z_s = \{x\}$ set theoretically.*

Proof. We may and do replace S by an affine open neighbourhood of s. We will prove the lemma for affine S by induction on $d = \dim_x(X_s)$.

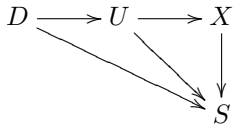
The case $d = 0$. In this case we show that we may take Z to be an open neighbourhood of x. (Note that an open immersion is a regular immersion.) Namely, if $d = 0$, then $X \to S$ is quasi-finite at x, see Morphisms, Lemma morphisms-lemma-locally-quasi-finite-rel-dimension-0. Hence there exists an affine open neighbourhood $U \subset X$ such that $U \to S$ is quasi-finite, see Morphisms, Lemma morphisms-lemma-quasi-finite-points-open. Thus after replacing X by U we see that the fibre X_s is a finite discrete set. Hence after replacing X by a further affine open neighbourhood of X we see that $f^{-1}(\{s\}) = \{x\}$ (because the topology on X_s is induced from the topology on X, see Schemes, Lemma schemes-lemma-fibre-topological). This proves the lemma in this case.

Next, assume $d > 0$. Note that because x is a closed point of its fibre the extension $\kappa(x)/\kappa(s)$ is finite (by the Hilbert Nullstellensatz, see Morphisms, Lemma morphisms-lemma-closed-point-fibre-locally-finite-type). Thus we see

$$\operatorname{depth}(\mathcal{O}_{X_s,x}) = \dim(\mathcal{O}_{X_s,x}) = d > 0$$

the first equality as $\mathcal{O}_{X_s,x}$ is Cohen-Macaulay and the second by Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point. Thus we may apply Lemma

lemma-slice-once to find a diagram



with $x \in D$. Note that $\mathcal{O}_{D_s,x} = \mathcal{O}_{X_s,x}/(\overline{h})$ for some nonzerodivisor $\overline{h}$, see Divisors, Lemma divisors-lemma-relative-Cartier. Hence $\mathcal{O}_{D_s,x}$ is Cohen-Macaulay of dimension one less than the dimension of $\mathcal{O}_{X_s,x}$, see Algebra, Lemma algebra-lemma-reformulate-CM for example. Thus the morphism $D \to S$ is flat, locally of finite presentation, and Cohen-Macaulay at x with $\dim_x(D_s) = \dim_x(X_s) - 1 = d - 1$. By induction hypothesis we can find a regular immersion $Z \to D$ having properties (a), (b), (c). As $Z \to D \to U$ are both regular immersions, we see that also $Z \to U$ is a regular immersion by Divisors, Lemma divisors-lemma-composition-regular-immersion. This finishes the proof. □

References

[1] Dieudonné, Jean; Grothendieck, Alexander. *Éléments de géométrie algébrique.* Inst. Hautes Études Sci. Publ. Math., volumes 4, 8, 11, 17, 20, 24, 28, 32, 1961–1967.